

BOF CONVERTER OPERATIONS MANUAL

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Equipment Description

The Basic Oxygen Furnace converts hot metal into steel by blowing oxygen through molten iron.

Normal Operating Parameters

- Steel Temperature: 1500°C - 1700°C
- Oxygen Flow: Within production recipe limits
- Vessel Tilt Operation: Normal cycle

Critical Thresholds

- Temperature below 1450°C
- Excessive oxygen deviation
- Hydraulic pressure loss

Common Failure Modes

1. Lance Failure
2. Hydraulic System Failure
3. Refractory Damage
4. Oxygen Flow Instability

Troubleshooting Guide

Low Steel Temperature

Possible Causes:

- Oxygen flow deviation
- Process interruption

Actions:

- Verify oxygen system
- Review heat process records

Hydraulic Failure

Possible Causes:

- Seal leakage
- Pump failure

Actions:

- Inspect hydraulic unit
- Verify pressure readings

Maintenance Recommendations

Daily:

- Check hydraulic pressures

Weekly:

- Inspect lance condition

Monthly:

- Refractory assessment